

UART error flags

A real UART RX does not just deliver bytes, it reports when something went wrong

Starter clean A5 eight-E-one

- Starter preset
- All three flags start at zero
- Byte A5 has four ones in its eight bits, so the expected even parity bit Pe is zero
- Step bit through idle, start, D0 through D7, Pe, stop, and trailing idle
- Play to end and the verdict should read RX OK with flags clear
- The RHR panel shows the latched byte unread until you click Read RHR

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

UART error cases

Spot classic RX faults on a short wave: **framing** (bad stop), **parity** (bad check bit), and **overrun** (new byte before the holding register is read). Starter: clean 0xA5 8E1 — no flags.

Starter example: clean frame 0xA5 with **even** parity and stop=1 — framing/parity/overrun all clear. Switch presets to inject bad stop, bad parity, or a second byte before RHR read.

Load starter example

Challenges 0 / 22 cleared

Quiz: framing: A framing error usually means...

☐ the sampled stop bit was not 1

☐ the start bit was 1

☐ parity matched

☐ the FIFO was empty

Show hint

Check

Next

Idle

Quiz: framing

Quiz: parity

Quiz: overrun

Quiz: break vs frame

Starter

Step once

Play clean

Framing preset

See framing

Land on bad stop

Parity preset

See parity err

Land on Pe

Overrun preset

See overrun

Read RHR

Demo

Explain

Clean odd

Reset cursor

A5 even Pe

Sketch flags

Core ideas

Framing

Stop bit must be 1. A 0 here usually means

Parity

Check bit must match even/odd over the data

Overrun

Hardware finishes a new byte while software

UART errors starter

learn_uart — uart errors

Real RTL/TB practice

- In Track A, name the three UART RX status flags and what each one means in one line
- For hex A5 eight-E-one, compute the expected even parity bit
- Sketch what software should do when framing is set versus parity only
- Optional: peek at UART status-register examples in the legacy combined materials
- This lab is error literacy, your TB should still inject and check these flags explicitly

Pitfalls to watch

- Framing and parity are sampled at the end of the frame
- A parity error can still latch data into RHR; framing often means the byte is suspect
- Overrun is a software-timing problem
- A line held low longer than a frame looks like repeated framing errors or break
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, list framing, parity, and overrun and one cause for each
- When you are ready, take the short quiz, then continue to FIFO in the datapath